EC39001: Analog Communication Lab Experiment 3: Amplitude Modulation using MC 1496

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1 Aim of the Experiment

To study and demonstrate amplitude modulation using the MC1496 balanced modulator IC by varying the message signal amplitude, and to observe under-modulation, critical modulation, and over-modulation in both time and frequency domains, including double sideband suppressed carrier (DSB-SC) characteristics.

2 Components Used

- Breadboard
- MC1496 IC
- Potentiometer
- Function Generator
- Oscilloscope
- Resistors: $1k\Omega$ (3), $4.7k\Omega$ (4), $18k\Omega$ (2), $3.3k\Omega$ (2), $100k\Omega$ (1), 820Ω (1)
- Capacitors: $22\mu F$ (1), $0.1\mu F$ (3), $0.47\mu F$ (1)
- DC Power Supply
- Connecting Wires

3 Theory

Amplitude modulation was performed using the MC1496, a balanced modulator-demodulator IC. By adjusting the message signal's amplitude, the three primary cases of AM—under-modulation, critical modulation, and over-modulation—were observed in both time and frequency domains.

The MC1496 is a 14-pin IC. Pins 1 and 4 are the differential inputs for the message (modulating) signal, while pins 8 and 10 are for the carrier input. The modulated output is taken from pins 6 and 12. Pin 5 is a bias input, and gain can be adjusted via a resistor connected between pins 2 and 3. Internally, the MC1496 functions as an analog multiplier. It uses an upper quad differential amplifier driven by a standard differential amplifier with dual current sources. This configuration results in a full-wave balanced multiplication of the two input signals, producing an output proportional to their product.

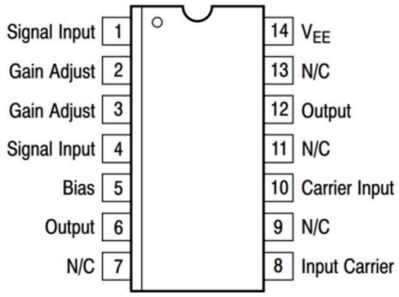


Figure 1: Pin Diagram of MC 1496

The modulation index (m) determines the type of modulation:

$$m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}} = \frac{A_{message}}{A_{carrier}}$$

- **Under-modulation ($m < 1$):** Occurs when the message signal amplitude is less than the carrier signal amplitude.
- **Critical modulation ($m = 1$):** Occurs when the message and carrier signal amplitudes are equal.
- **Over-modulation ($m > 1$):** Occurs when the message signal amplitude is greater than the carrier amplitude, causing distortion.

4 Circuit Diagram

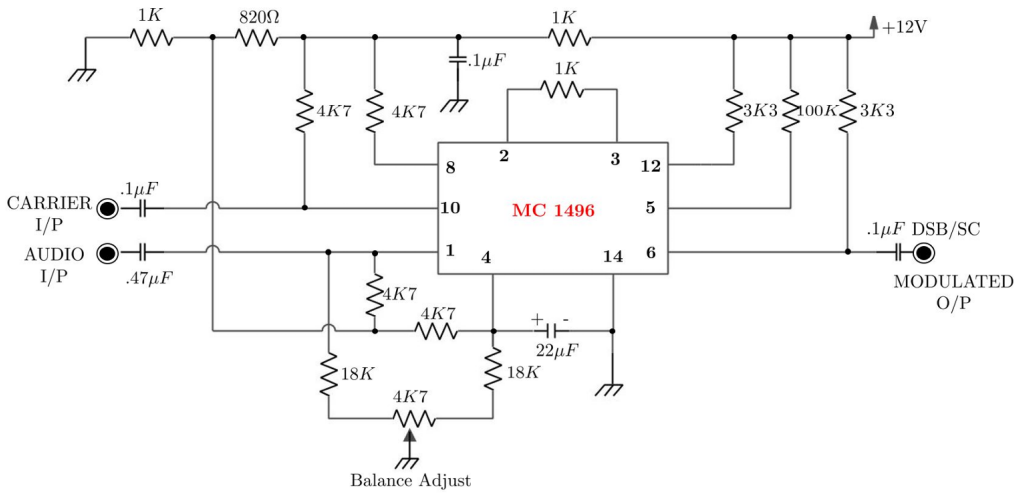


Figure 1: Circuit diagram

Figure 2: Circuit Diagram for AM using MC 1496

5 Observations and Results

The modulation index (m) was calculated from the oscilloscope waveform using the formula:

$$m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

5.1 Under-Modulation

Input signals were set to $f_m = 2kHz$, $f_c = 20kHz$, $A_m = 150mV_{p-p}$, and $A_c = 300mV_{p-p}$.

- **Theoretical Modulation Index:** $m = A_m/A_c = 150/300 = 0.5$.
- **Measured Modulation Index:** With $A_{max} = 336\,mV$ and $A_{min} = 124\,mV$, the calculated index was $m \approx 0.46$.

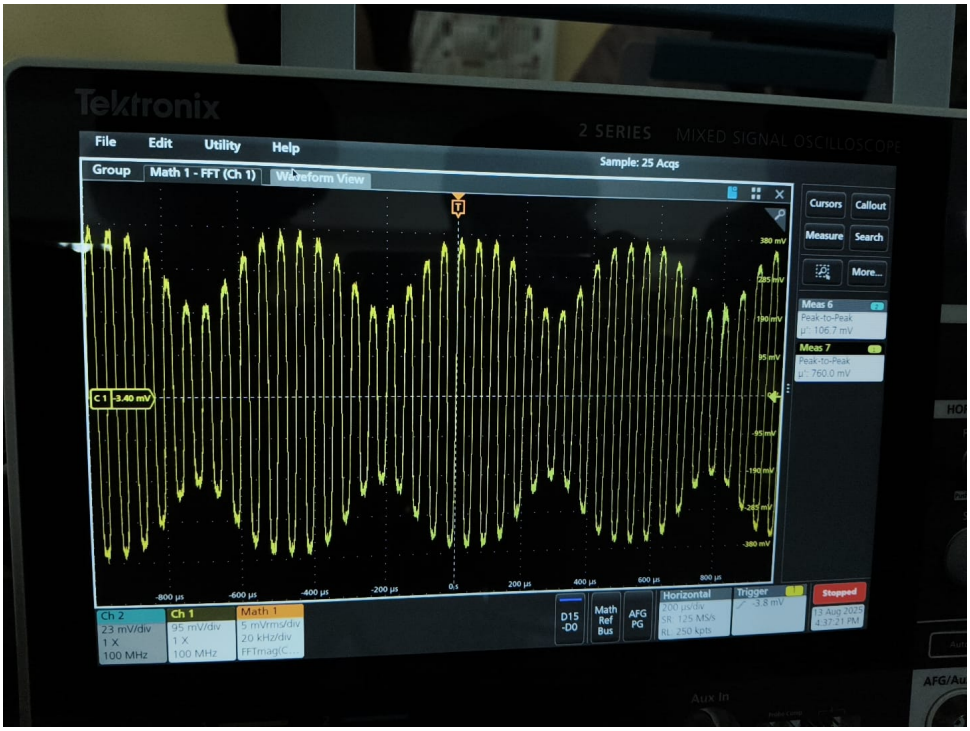


Figure 3: Undermodulated Wave (Time Domain)

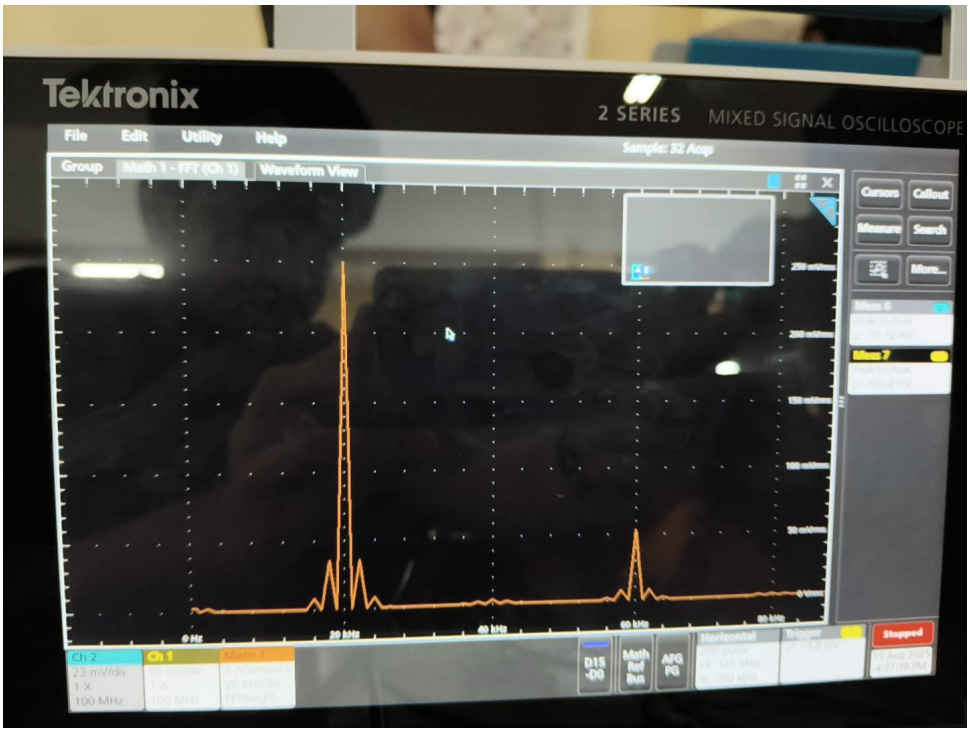


Figure 4: FFT of Undermodulated Wave (Unsuppressed Carrier)

Figure 4 shows the FFT of Undermodulated Wave (Unsuppresed Carrier) . We can observe three peaks carrier peak at 20 kHz , and Sideband at 18 kHz and 22 kHz. Peak of carrier overshadowing that of message signal.

5.2 Critical Modulation

Input signals were set to $f_m = 2kHz$, $f_c = 20kHz$, $A_m = 300mV_{p-p}$, and $A_c = 300mV_{p-p}$.

- **Theoretical Modulation Index:** $m = A_m/A_c = 300/300 = 1$.
- **Measured Modulation Index:** With measured values, the calculated index was $m \approx 0.914$.

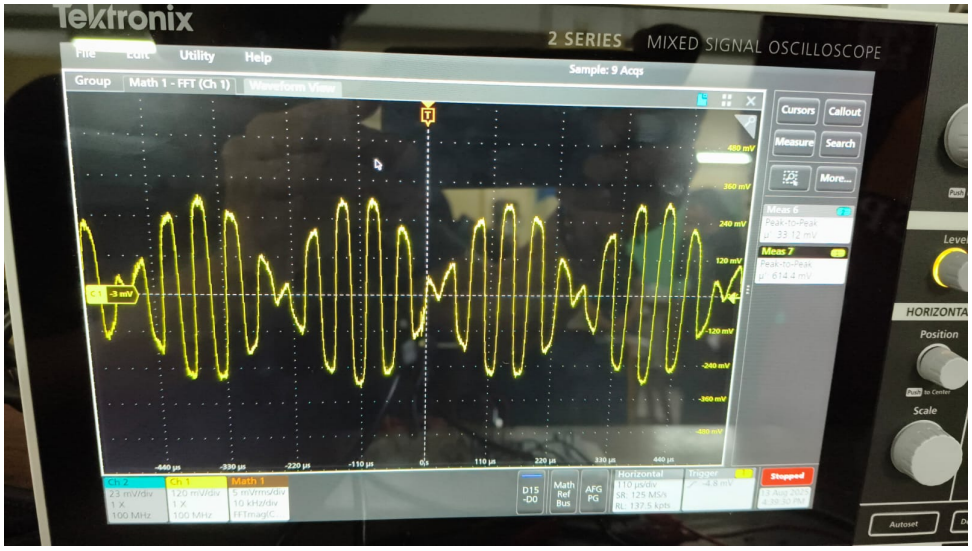


Figure 5: Critically Modulated Wave (Time Domain)

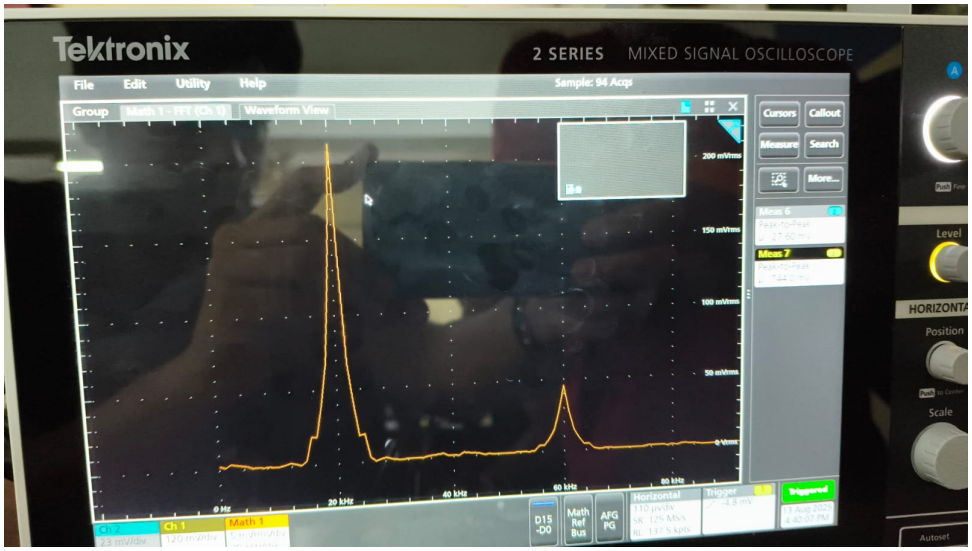


Figure 6: FFT of Criticalaly Modulated Wave (Unsuppressed Carrier)

Figure 6 shows the FFT of Criticaly Modulated Wave (Unsuppresed Carrier) . Frequency peaks seems to merge , it is due to limitation of Digital Oscilloscope.We can observe three peaks carrier peak at 20 kHz , and Sideband at 18 kHz and 22 kHz. Peak of carrier overshadowing that of message signal.

5.3 Over-Modulation

Input signals were set to $f_m = 2kHz$, $f_c = 20kHz$, $A_m = 300mV_{p-p}$, and $A_c = 150mV_{p-p}$.

- **Theoretical Modulation Index:** $m = A_m/A_c = 300/150 = 2$.
- The observed waveform showed significant envelope distortion, characteristic of over-modulation.

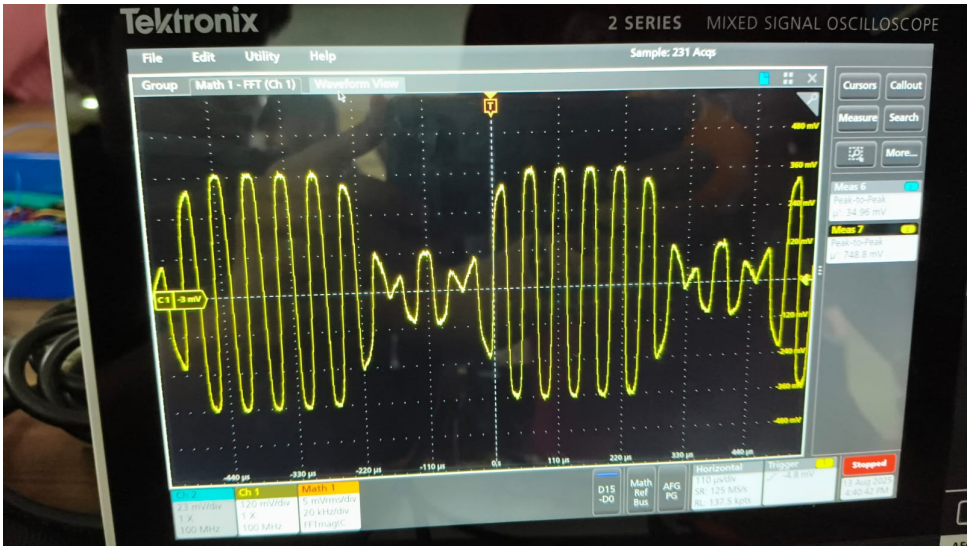


Figure 7: Overmodulated Wave (Time Domain)

Figure 7 shows the over-modulated waveform for $f_m = 2\,\text{kHz}$, $f_c = 20\,\text{kHz}$, $A_m = 300\,\text{mV}$ (p-p), and $A_c = 150\,\text{mV}$ (p-p). Theoretically, the modulation index is 2. From the obtained waveform, the calculated modulation index is also close to 2.



Figure 8: FFT of Overmodulated Wave (Unsuppressed Carrier)

Figure 8 shows the FFT of the over-modulated wave. Unlike the under-modulated case, additional peaks appear at higher frequencies.

5.4 Double Sideband-Suppressed Carrier (DSB-SC)

By carefully adjusting the "Balance Adjust" potentiometer, the carrier component was suppressed from the output. The resulting frequency spectrum showed peaks only at the sideband frequencies ($f_c \pm f_m$), with the carrier peak at f_c significantly attenuated.

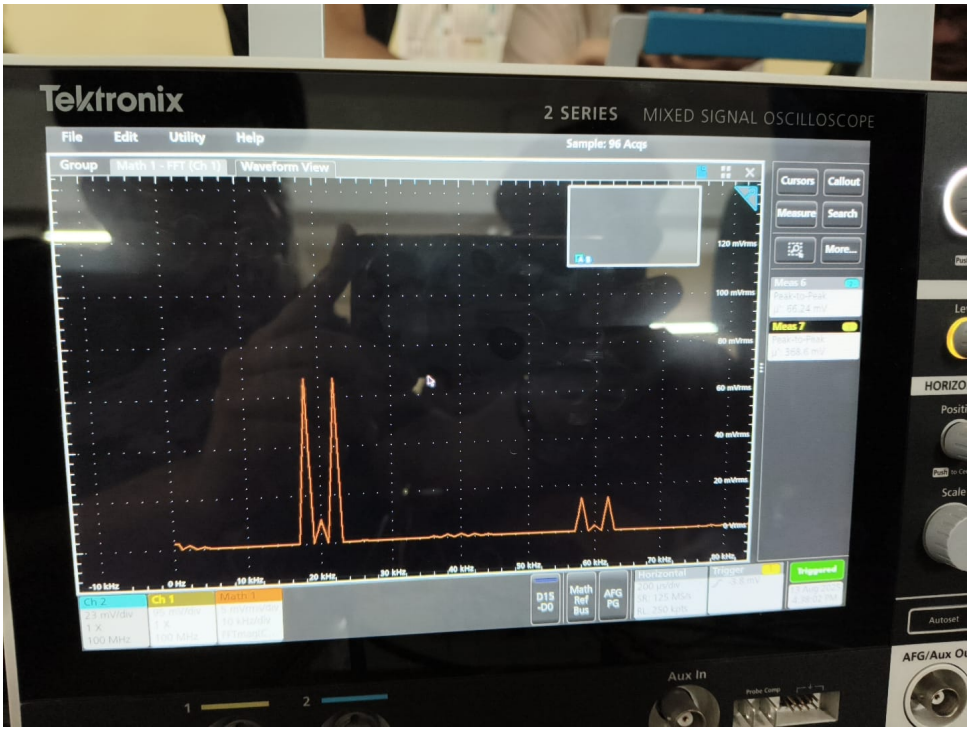


Figure 9: DSB-SC Spectrum of Undermodulated Signal (Carrier Suppressed)

Figure 9 presents the FFT of the under-modulated waveform after suppressing the carrier. Peaks are seen only at the sum and difference frequencies, i.e., 22 kHz and 18 kHz.



Figure 10: DSB-SC Spectrum of Overmodulated Siganal (Carrier Suppressed)

Figure 10 presents the FFT of the over-modulated waveform with the carrier suppressed.



Figure 11: DSB-SC Spectrum of Criticaly Modulated Signal (Carrier Suppressed)

Figure 11 shows the FFT of the critically-modulated waveform with the carrier suppressed.

6 Discussion

The experiment provided a comprehensive practical study of amplitude modulation using the MC1496 integrated circuit, which functions as a four-quadrant analog multiplier. The results obtained were largely in agreement with theoretical principles, and the minor deviations observed can be attributed to practical circuit non-idealities.

7 Discussion

This experiment showed how to create Amplitude Modulation (AM) signals using the MC1496 chip. The chip works by multiplying the audio and carrier signals. Our results in the lab were very close to what the theory says they should be. The small differences we saw were because real-life electronic parts are not perfect.

- **Checking the Modulation Index (m):** The main goal was to see what happens when we change the modulation index, which is called 'm'. We looked at three different cases.
  - **Under-Modulation ($m < 1$):** In this test, the shape of the final wave correctly matched the original audio signal, so there was no distortion. The frequency graph showed a big peak for the carrier at 20 kHz and two smaller side-peaks at 18 kHz and 22 kHz. We calculated our m to be 0.46, which is very close to the planned value of 0.5.

  - **Critical Modulation ($m = 1$):** Here, we tried to make m exactly equal to 1. Our measurement was about 0.914. This is the best you can do for regular AM because it makes the signal as strong as possible without messing up its shape.
  - **Over-Modulation ($m > 1$):** When we made the audio signal too strong, we got over-modulation. This caused the final wave's shape to be distorted. On the frequency graph, we saw extra, unwanted peaks appear. This is a problem because in real life, it would cause interference with other radio stations.

- **Creating a DSB-SC Signal:** The MC1496 chip can also work as a balanced modulator, which lets us remove the carrier signal. We did this by turning the "Balance Adjust" knob. This created a Double Sideband-Suppressed Carrier (DSB-SC) signal. On the frequency graph, the big peak for the carrier was gone, and only the two sideband peaks were left. This is useful because it saves a lot of power by not transmitting the carrier, which doesn't contain any of the audio information.

- **Overall Performance:** In the end, the experiment proved that the MC1496 chip works very well for making AM signals. Everything we saw on our equipment matched the theory. We learned how the chip can make both regular AM and the power-saving DSB-SC type of signal.

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